

- Track preclinical and clinical trials
- Link to specific vaccine batches/candidates
- Document outcomes and regulatory submissions
- Timeline visualization

6. Equipment Integration:

- Bi-directional for Real-Time PCR, ELISA Reader, Luminex, Automated Cell Counter
- Manual upload for Nanodrop, Qubit, Gel Electrophoresis, SDS-PAGE
- Environmental monitoring for Refrigerators, Freezers, Incubators

TRADITIONAL & MODERN MEDICINE RESEARCH & DEVELOPMENT (TMMRD) LABORATORY - DATA CONFIGURATION SUMMARY

LABORATORY OVERVIEW

TMMRD Laboratory focuses on traditional medicinal plants, extraction, phytochemical analysis, safety/efficacy testing, and formulation of herbal medical products. Combines traditional knowledge with modern analytical techniques.

1. SAMPLE TYPES

Plant Materials:

- Whole plants
- Plant parts: Root, stem, leaf, flower, seed, bark, fruit
- Fresh plant materials
- Dried plant materials
- Herbarium specimens

Processed Materials:

- Plant extracts (aqueous, ethanolic, methanolic, etc.)

- Fractions (after chromatographic separation)
 - Purified compounds (isolated active constituents)
 - Formulated products (capsules, tinctures, ointments, teas)
-

2. KEY METADATA FIELDS

Sample Arrival:

- Sample ID (unique identifier)
- Plant species (scientific name, common name)
- Plant part (root, leaf, etc.)
- Origin (geographic location, collection site)
- Collector (person/organization)
- Collection date & time
- Fresh vs. dried status
- Authentication status (botanically verified or pending)

Processing Metadata:

- Processing method (drying, grinding, chopping, powdering)
- Extraction solvent and method
- Filtration/concentration method
- Yields (weight or volume at each step)
- Date & time
- Operator

Product Development Metadata:

- Formulation type (capsule, tincture, ointment, tea)
 - Ingredients and quantities
 - Manufacturing date
 - Batch number
 - Stability data
-

3. COMPLETE WORKFLOW STAGES

STAGE 1: Sample Intake & Registration

Sample Arrival:

- Plant materials or traditional medicine samples received
- May arrive fresh or dried

Registration & Labeling:

- Assign unique sample ID
 - Record metadata (origin, species, collector, date/time)
-

STAGE 2: Authentication

Process:

- Botanical verification or expert identification
- Methods:
 - Morphological examination
 - Microscopic analysis
 - Molecular identification (DNA barcoding: ITS, rbcL, matK genes)

Documentation:

- Authentication method used
 - Result (species confirmed, uncertain, misidentified)
 - Authenticator name and date
 - If herbarium specimen prepared, link to herbarium catalog number
-

STAGE 3: Sample Storage & Herbarium Placement

Storage Options:

- **Fresh samples:** Refrigerated (2-8°C) for short-term
- **Dried samples:** Room temperature in sealed containers (desiccant if needed)
- **Preserved samples:** In fixative or alcohol

Herbarium Placement (for reference specimens):

- Mount, label, and catalog specimen
- Record herbarium entry:
 - Specimen ID
 - Species, collector, location, date
 - Herbarium cabinet location
- Link to LMIS sample record

Link to Projects:

- Herbarium entries linked to ongoing analytical or formulation projects
-

STAGE 4: Sample Preparation for Analysis

Physical Processing:

- **Freshly processed samples:** Used immediately
- **Samples needing drying:**
 - Air drying
 - Oven drying (controlled temperature)
 - Freeze drying (for heat-sensitive materials)

Preparation Methods:

- Grinding (to powder)
- Chopping (coarse pieces)
- Powdering (fine powder for extraction)

Documentation:

- Processing method
 - Weight before/after processing
 - Yield (% of starting material)
 - Date, operator
-

STAGE 5: Extraction, Filtration, & Concentration

Extraction Process:

- **Solvents:** Ethanol, methanol, water, hexane, chloroform, etc.
- **Techniques:**
 - Maceration (soaking)
 - Soxhlet extraction
 - Ultrasonic extraction
 - Distillation (for essential oils)
 - Others

Documentation:

- Solvent type and volume

- Plant material weight
- Extraction technique
- Temperature and duration
- Yield (extract weight or volume)
- Operator, date

Filtration:

- Remove plant debris and impurities
- Method: Filter paper, vacuum filtration, centrifug

ation

Concentration:

- Evaporation (rotary evaporator)
- Distillation
- Reduce volume to enrich extract

Record:

- Final extract volume/weight
- Yield (% extraction efficiency)

STAGE 6: Analytical Pathways

Two Pathways:

PATH A: Advanced Analysis (Before Production)

If further profiling required:

Fractionation:

- Separate extract into fractions using chromatography (column chromatography, HPLC prep)
- Record method and fractions collected

Identification/Isolation:

- Detect active constituents
- Techniques: TLC, HPLC, GC-MS, LC-MS
- Link to compound database (if known compounds)

Purification:

- Remove impurities to isolate pure compound
- Methods: Recrystallization, prep-HPLC, column chromatography
- Record purity level (% , assessed by HPLC or NMR)

Characterization:

- Determine structure and properties
- Techniques:
 - NMR (Nuclear Magnetic Resonance)
 - Mass Spectrometry (MS)
 - Infrared Spectroscopy (IR/FTIR)
- Store spectral data (files linked to compound record)

Proceed to Product Development

PATH B: Direct to Production

If extract ready post-concentration:

- Skip advanced analysis
- Proceed directly to product development

STAGE 7: Product Development & Testing

Preliminary Phytochemical Screening:

- Test for presence of compound classes:
 - Alkaloids
 - Flavonoids
 - Tannins
 - Saponins
 - Terpenoids
 - Glycosides
 - Others
- Record test results (positive/negative for each class)

Safety/Toxicity Study:

- **In vitro:** Cell-based toxicity assays
- **In vivo:** Animal models (if applicable)
- Link to external trial data (if conducted elsewhere)
- Record outcomes:
 - LD50 (if determined)
 - No Observed Adverse Effect Level (NOAEL)

- Safety classification

Efficacy Test:

- **Biological activity assays:**
 - Antimicrobial (against bacteria, fungi)
 - Antioxidant (DPPH, ABTS assays)
 - Anti-inflammatory
 - Anticancer (cell viability assays)
 - Others (depending on traditional use)
 - Document outcomes and protocols
-

STAGE 8: Formulation of Medical Product

Formulation Types:

- Capsules
- Tinctures (alcohol-based extracts)
- Ointments/creams (topical)
- Teas (herbal infusions)
- Syrups
- Others

Formulation Process:

- Select formulation based on intended use and stability
- Record:
 - Ingredients (extract + excipients/carriers)
 - Quantities/concentrations
 - Manufacturing steps
 - Batch number
 - Manufacturing date

Product Testing:

- Stability testing (accelerated and real-time)
- Microbial contamination testing
- Heavy metal testing
- Pesticide residue testing (if applicable)
- Active constituent quantification

Documentation:

- Product specifications

- Batch records
 - QC results
-

4. ASSAY/TEST CATALOG

Botanical/Authentication:

- Morphological examination
- Microscopic analysis
- DNA barcoding (ITS, rbcL, matK)

Extraction:

- Various solvent extractions
- Distillation (essential oils)

Phytochemical Screening:

- Alkaloid test
- Flavonoid test
- Tannin test
- Saponin test
- Terpenoid test
- Glycoside test
- Others

Analytical Techniques:

- TLC (Thin Layer Chromatography)
- HPLC (fingerprinting, quantification)
- GC-MS (volatile compounds)
- LC-MS (compound identification)
- NMR (structure elucidation)
- IR/FTIR (functional groups)

Biological Activity Assays:

- Antimicrobial (disc diffusion, MIC)
- Antioxidant (DPPH, ABTS, FRAP)
- Anti-inflammatory (in vitro enzyme inhibition, in vivo models)
- Anticancer (MTT, SRB, cell viability assays)
- Others

Safety/Toxicity:

- Acute toxicity (LD50)
- Cytotoxicity assays
- Animal models (if applicable)

Product QC:

- Stability testing
 - Microbial testing (TAMC, TYMC, pathogens)
 - Heavy metals (Pb, Cd, Hg, As)
 - Pesticide residues
 - Active constituent quantification
-

5. QUALITY CONTROL PARAMETERS

Plant Material QC:

- Correct species (authenticated)
- Free from contamination (microbial, pesticides)
- Adequate condition (not spoiled, moldy)

Extraction QC:

- Yield within expected range
- Extract quality (color, odor, consistency)
- No contamination

Analytical QC:

- Controls and standards run with each assay
- Instrument calibration current
- Method validation (if new method)

Product QC:

- Meets specifications for:
 - Active constituent content
 - Microbial limits
 - Heavy metals
 - Stability
- Pass/Fail based on pharmacopeial standards or in-house specs

Acceptance Criteria:

- All QC tests pass
 - Product safe and efficacious
-

6. DATA HANDLING REQUIREMENTS

Data Capture:

- LMIS for sample tracking, authentication, extraction, testing
- Analytical instrument outputs (HPLC chromatograms, spectra)
- Images (herbarium specimens, TLC plates, microscopy)

Data Processing:

- Chromatographic data analysis (peak identification, integration)
- Spectral interpretation
- Statistical analysis of bioactivity results

ALCOA+ Compliance:

- Standard compliance principles

Review Workflow:

1. Primary Review (Lab Technician/Researcher)
 2. Secondary Review (Senior Scientist/Pharmacognosist)
 3. Final Approval (Lab Manager/Principal Investigator)
-

7. STORAGE SPECIFICATIONS

Hierarchy:

- **Herbarium:** Cabinet → Shelf → Specimen folder
- **Plant Materials:** Room → Storage cabinet → Container
- **Extracts/Products:** Room → Refrigerator/Freezer → Shelf → Box → Vial

Storage Conditions:

- **Fresh plant materials:** Refrigerator 2-8°C (short-term)
- **Dried plant materials:** Room temperature, dry, dark
- **Extracts:** Refrigerator (2-8°C) or freezer (-20°C) depending on stability
- **Purified compounds:** Refrigerator or freezer, protected from light

- **Formulated products:** Per product specification (often room temp or refrigerated)
- **Herbarium specimens:** Room temperature, low humidity, protected from pests

Environmental Monitoring:

- Temperature and humidity monitoring
 - Pest control in herbarium
-

8. DISPOSAL & ARCHIVING

Disposal Criteria:

- Sample exhausted
- Failed QC (contaminated, degraded)
- Retention period expired

Disposal Methods:

- **Plant materials:** Composting or disposal as biodegradable waste
- **Extracts/chemicals:** Chemical waste disposal per institutional guidelines
- **Contaminated materials:** Autoclaving (if microbial contamination)

Disposal Documentation:

- Standard logging with approvals

Herbarium Archiving:

- Specimens retained indefinitely (reference collection)
- Cataloged and accessible for future research

Data Archiving:

- Analytical data, spectra, reports: Permanent retention
 - Product batch records: Per regulatory requirements (typically 5-10 years)
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9. PERFORMANCE METRICS & REPORTING

Routine Reports:

- Sample intake logs (plant materials received)

- Authentication success rates
- Extraction yields (average per species/part)
- Product batch production logs

Quality Metrics:

- Contamination rates (plant materials, extracts)
- QC pass rates (extracts, products)
- Bioactivity success rates (% showing activity)

Dashboards:

- Herbarium inventory (species count, specimens)
- Extraction pipeline status
- Product development stage tracking

External Reporting:

- Collaborators (analytical results)
- Regulatory agencies (product registration)
- Publications (phytochemical profiles, bioactivity data)

10. ANALYZERS/MACHINES

From Other Lab Lists (Shared Equipment):

- HPLC (from Bioanalytical/Pharma)
- LC-MS (from Bioanalytical/Pharma)
- GC-MS (if available)
- FTIR (from Pharma)
- UV-Vis Spectrophotometer (from Pharma)
- TLC equipment
- Rotary evaporator
- Freeze dryer
- Microscopes (optical, fluorescence)

Note: TMMRD likely shares equipment with Pharmaceutical/Bioanalytical labs. No dedicated machine list provided in SRS.

11. USER ROLES & PERMISSIONS

Role	Registration	Authentication	Processing	Extraction	Analysis	Product Development	Approval	Herbarium
Lab Technicians	Yes	No	Yes	Yes	Yes	Yes	No	View
Researchers	Yes	No	Yes	Yes	Yes	Yes	No	View
Pharmacognosists	View	Yes	View	View	View	Yes	Approve	Full
Lab Manager	Full	Full	Full	Full	Full	Full	Final Approve	Full
Principal Investigator	View	View	View	View	View	View	Final Approve	View

12. KEY SYSTEM REQUIREMENTS

Essential Features:

1. Plant Species Management:

- Botanical name (genus, species, authority)
- Common names (multiple languages)
- Synonyms
- Link to external databases (e.g., The Plant List, GBIF)

2. Authentication Workflow:

- Record authentication method
- Link to authenticator (botanist/expert)
- Store authentication evidence (photos, DNA sequences)

3. Herbarium Management:

- Specimen catalog with location tracking
- Link physical specimens to LMIS records
- Search by species, collector, location, date
- Image storage (specimen photos)

4. Extraction Yield Calculation:

- Auto-calculate % yield from input/output weights
- Track yields across batches (for optimization)

5. Dual Pathway Support:

- Advanced Analysis path (Fractionation → Identification → Purification → Characterization)
- Direct to Production path
- User selects path at appropriate stage

6. Phytochemical Screening Results:

- Checklist for compound classes (Alkaloids, Flavonoids, etc.)
- Positive/Negative for each
- Semi-quantitative results (if applicable: +, ++, +++)

7. Product Formulation Records:

- Formulation type selection (capsule, tincture, etc.)
- Ingredient list with quantities
- Batch number generation
- Manufacturing date
- QC results linkage

8. Bioactivity Data Entry:

- Assay type (antimicrobial, antioxidant, etc.)
- Organism/target (if antimicrobial)
- IC50, MIC, % inhibition, or other metrics
- Positive controls
- Interpretation (active, inactive, moderate activity)

9. Safety/Toxicity Tracking:

- Link to external trial data (if applicable)
- Record LD50, NOAEL, safety classification

10. Analytical Data Storage:

- Upload chromatograms (HPLC, GC-MS)
- Upload spectra (NMR, IR, MS)
- Link to samples/compounds
- Annotation support (peak identification)

11. Inventory Tracking:

- Track extracts, fractions, purified compounds
- Volume/weight remaining
- Storage location

- Expiry tracking

12. Product Batch QC:

- Record stability data (time points, conditions, results)
- Microbial testing results
- Heavy metal testing results
- Active constituent quantification

BIOREPOSITORY LABORATORY - DATA CONFIGURATION SUMMARY

ETA for UAT Release: January 16th

SRS Functional Reqs: [AHRI - Software Requirements Specification \(SRS\)](#)

1. SAMPLE TYPES

All Sample Types from AHRI Labs:

- Biological samples (serum, plasma, DNA, RNA, tissue, cells, isolates)
- Non-biological samples (pharmaceuticals, reagents, reference materials)
- Project-concluded samples from all laboratory units
- External collaborator samples (with appropriate documentation)

2. KEY METADATA FIELDS

Intake Metadata:

- Originating laboratory or external source
- Project/study association and Principal Investigator
- Date, time, and location of collection and receipt
- Sample type and category